

IntelliVault - AI Powered Smart Knowledge Management System

Objective

Build a full-stack project integrating DBMS, AI and a modern UI.

Technology Stack

Frontend: HTML5, CSS3, Bootstrap 5, JavaScript

Backend: Python + Flask

Database: MySQL

AI: Python, LLM API/Local Model, NLP

Core Modules

1. Authentication (Login/Signup)
2. Dashboard
3. Notes CRUD
4. PDF Upload
5. AI Summarization
6. Keyword Extraction
7. AI Chat with Notes
8. Smart Search
9. Quiz & Flashcard Generator
10. Profile & Settings

Database Tables

Users(id,name,email,password)

Notes(id,user_id,title,content,date)

Summaries(id,note_id,summary)

Keywords(id,note_id,keyword)

Quiz(id,question,option1,option2,option3,option4,answer)

Folder Structure

app.py, templates/, static/, database/, AI/, uploads/

Development Roadmap

Phase 1: Project setup, Flask, Bootstrap, MySQL.

Phase 2: Authentication and Notes CRUD.

Phase 3: PDF upload, AI summary, keywords.

Phase 4: AI chat, semantic search, quiz, flashcards.

Phase 5: Testing, optimization, deployment.

Future Enhancements

Role-based access, analytics, cloud deployment, notifications, OCR, multilingual support.